

# Particle Swarm Optimization and the Fully Informed Particle Swarm

## A Comparative Analysis

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- 1 Introduction
- 2 Optimization and Swarm Intelligence
- 3 Canonical PSO
- 4 PSO vs. FIPS
- 5 Numerical Simulations
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*“That which is not good for the swarm, neither is it good for the bee.”*  
— Marcus Aurelius, *Meditations*

- Identifying the optimal configuration of a system is fundamentally a **search problem**.
- Classical derivative-based methods fail when the objective function is non-differentiable, discontinuous, or analytically unavailable.
- **PSO** (Kennedy & Eberhart, 1995) is a derivative-free metaheuristic in which a population of interacting particles collectively explores the search space.

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## 5 principles of swarm systems (*Millonas*)

- Proximity, Quality, Diverse response
- **Stability vs. Adaptability**

This stability/adaptability trade-off anticipates the comparison between PSO and FIPS developed in Chapter 4.

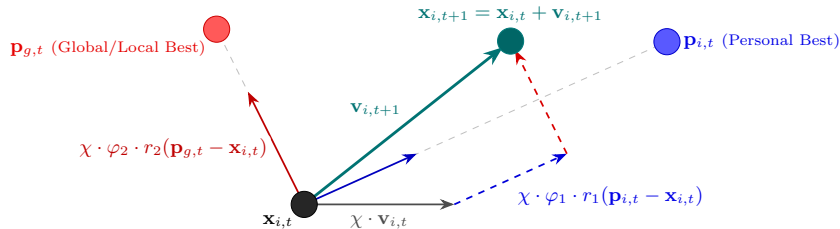
## Two direct sources of PSO

- **Boid model** (Reynolds, 1987)
- **Cornfield model** (Heppner)
- Kennedy & Eberhart (1995) formalized these ideas into the original PSO update rule.

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# Velocity and Position Update



Vector decomposition of the update (adapted from the thesis, Fig. 3.1); here  $\chi$  replaces  $\omega$  and  $\varphi_1, \varphi_2$  replace  $c_1, c_2$  in the constriction-factor formulation used throughout this thesis:

$$v_i^d(t+1) = \chi \left[ v_i^d(t) + \varphi_1 r_1 (p_i^d - x_i^d(t)) + \varphi_2 r_2 (p_g^d - x_i^d(t)) \right]$$



- **Cognitive term** ( $\varphi_1$ ): attracts the particle toward its own personal best  $p_i$ , representing its individual experience.
- **Social term** ( $\varphi_2$ ): attracts the particle toward the swarm's best  $p_g$ , representing social cooperation.
- Higher  $\varphi_1/\varphi_2 \Rightarrow$  conservative behavior, with greater emphasis on the particle's own experience.
- Higher  $\varphi_2/\varphi_1 \Rightarrow$  faster but riskier convergence toward the swarm best.

Historical evolution: fixed coefficients (Kennedy & Eberhart, 1995)  $\rightarrow$  inertia weight  $\omega$  (Shi & Eberhart, 1998)  $\rightarrow$  constriction factor  $\chi$  (Clerc & Kennedy, 2002).

# The Constriction Factor

- Controls the dynamics of particle trajectories and provides stability without the need for explicit velocity clamping (*Clerc & Kennedy, 2002*).
- It is derived from a stability analysis of the swarm as a discrete-time linear dynamical system, with the eigenvalues constrained within the unit circle.

$$\chi = \begin{cases} \frac{2\kappa}{\left| 2 - \varphi - \sqrt{\varphi^2 - 4\varphi} \right|} & \varphi = \varphi_1 + \varphi_2 > 4 \\ \kappa & \text{otherwise} \end{cases}$$

- **Thesis setting:**  $\kappa = 1$ ,  $\varphi_1 = \varphi_2 = 2.05 \Rightarrow \varphi = 4.1$ ,  $\chi \approx 0.7298$ .

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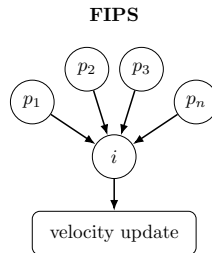
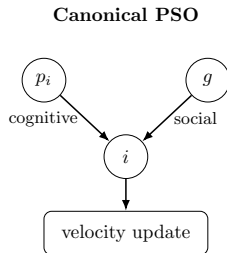
# From PSO to FIPS: Motivation

## Limitation of canonical PSO

The social component is based on a *single* social reference (the best-performing neighbor) discarding the experience of every other particle in the neighborhood.

## Fully Informed Particle Swarm (*Mendes, Kennedy & Neves, 2004*)

Every neighbor contributes to the velocity update, not just the best-performing one.



## FIPS: Formulation (Unweighted, Self-Inclusive)

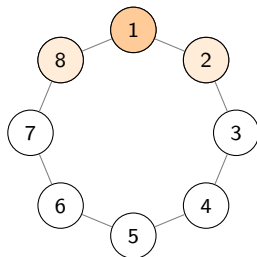
$$v_{i,t+1}^d = \chi \left( v_{i,t}^d + \varphi \left( p_{m,i,t}^d - x_{i,t}^d \right) \right), \quad p_{m,i,t}^d = \frac{\sum_{k \in N_i} W(k) \varphi_k^d p_{k,t}^d}{\sum_{k \in N_i} W(k) \varphi_k^d}$$

$$\varphi_k^d = U \left[ 0, \frac{\varphi}{|N_i|} \right] \quad \forall k \in N_i$$

- $N_i$ : neighborhood of particle  $i$  (here **including**  $i$  itself)
- $p_k$ : personal best of neighbor  $k$ ; unweighted  $\Rightarrow W(k) = 1$  for all  $k$
- **Thesis setting:**  $\varphi = 4.1$  (same total coefficient as canonical PSO),  $\chi \approx 0.7298$
- Canonical PSO = special case with  $|N_i| = 2$ , weighted by fixed  $\varphi_1, \varphi_2$

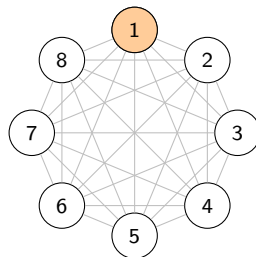
# Neighborhood Topologies

**Ring ( $K = 3$ )**



Each particle is connected only to its two immediate neighbors.

**All ( $K = N$ )**



Every particle is connected to every other particle.

# PSO vs. FIPS: Analogies and Differences

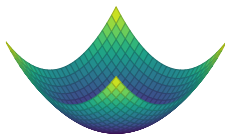
|                               | Canonical PSO                               | FIPS                                               |
|-------------------------------|---------------------------------------------|----------------------------------------------------|
| Social information source     | Single best neighbor                        | Weighted average of all neighbors                  |
| Effect of larger neighborhood | More candidates from which to pick the best | More (potentially conflicting) contributions       |
| Weighting function $W(k)$     | Not applicable                              | Determines the relative influence of each neighbor |
| Convergence mechanism         | Constriction factor $\chi$                  | Same $\chi$ , generalized to $K$ sources           |

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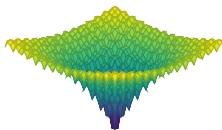
# Benchmark Functions

Sphere



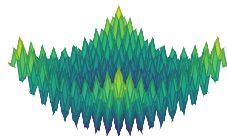
Unimodal, convex

Ackley



Nearly flat, deep central basin,  
many small local optima

Rastrigin



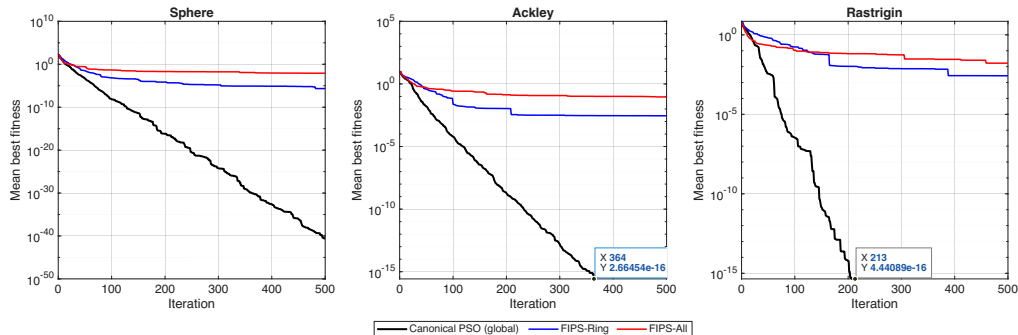
Highly multimodal, regularly spaced  
local optima

All three functions have global minimum  $f(0,0) = 0$ . Actual search domains: Sphere  $[-100, 100]^2$ , Ackley  $[-32, 32]^2$ , Rastrigin  $[-5.12, 5.12]^2$  (plots above use a common scale for visual comparison).

| Parameter                               | Value                      |
|-----------------------------------------|----------------------------|
| Swarm size $N$                          | 30                         |
| Dimensionality $D$                      | 2                          |
| Independent runs                        | 20 per configuration       |
| Max. iterations                         | 500                        |
| Boundary handling                       | Reflecting wall            |
| Success tolerance $\delta$              | $0.0015 \times (ub - lb)$  |
| Tight tolerance $\delta_{\text{tight}}$ | $10^{-4} \times (ub - lb)$ |

Same random initial swarm ( $X_0, V_0$ ) shared across all three configurations, per run. Tolerances follow Lange, Schmitt & Wanka (2014), calibrated for low-dimensional search spaces ( $D = 2$ ) and scaled to the domain width.

# Results: Convergence Behavior



- Canonical PSO decreases sharply and almost monotonically on all three functions, reaching machine precision well before iteration 500.
- Both FIPS configurations decrease for  $\sim 30$ – $50$  iterations, then **flatten out** — an early qualitative indication of stagnation.

## Results: Final Fitness, Success Rate & Stagnation

| Function  | Config.   | Mean best              | SR ( $\delta$ ) | SR ( $\delta_{\text{tight}}$ ) | Med. iters ( $\delta_{\text{tight}}$ ) |
|-----------|-----------|------------------------|-----------------|--------------------------------|----------------------------------------|
| Sphere    | Can. PSO  | $2.63 \times 10^{-41}$ | 100%            | 100%                           | 44.5                                   |
|           | FIPS-Ring | $2.18 \times 10^{-6}$  | 100%            | 100%                           | 80.0                                   |
|           | FIPS-All  | $8.16 \times 10^{-3}$  | 100%            | 20%                            | 363.5                                  |
| Ackley    | Can. PSO  | $\approx 0$            | 100%            | 100%                           | 47.0                                   |
|           | FIPS-Ring | $2.86 \times 10^{-3}$  | 100%            | 95%                            | 70.0                                   |
|           | FIPS-All  | $9.12 \times 10^{-2}$  | 100%            | 20%                            | 82.0                                   |
| Rastrigin | Can. PSO  | 0                      | 100%            | 100%                           | 55.5                                   |
|           | FIPS-Ring | $2.67 \times 10^{-3}$  | 100%            | 85%                            | 185.0                                  |
|           | FIPS-All  | $1.66 \times 10^{-2}$  | 95%             | 35%                            | 122.0                                  |

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# The Core Argument, in Three Sentences

- ① More information sources do not necessarily lead to a better-performing swarm.
- ② Neighborhood topology is not merely a technical detail—it affects the resulting search dynamics.
- ③ Similar fitness values can correspond to substantially different positions: therefore, an appropriate convergence criterion is essential.

## Limitations

- Restricted to  $D = 2$
- Only unweighted FIPS, Ring & All topologies
- Symmetric initialization (vs. Mendes' asymmetric design)

## Future work

- Extend the analysis to higher dimensionality
- Investigate weighted FIPS variants & more topologies
- Introduce formal statistical significance tests

**Thank you for your attention**



# Backup Slides

for additional questions

Constriction factor: full derivation

Full results table

5 topologies

FIPS variants

All limitations

# Backup: Constriction Factor — Derivation (Clerc & Kennedy, 2002, §IV-C)

Model (Eq. 4.8–4.9):  $v(t+1) = \chi(v(t) + \varphi(p - x(t)))$ ,  $x(t+1) = v(t+1) + x(t)$ .

**Step 1) Matrix of the System.**

$$M = \begin{pmatrix} \chi & \chi\varphi \\ -\chi & 1 - \chi\varphi \end{pmatrix}$$

**Step 2) Eigenvalues.** Roots of  $Z^2 - \text{tr}(M)Z + \det(M) = 0$ :

$$Z^2 - (\chi + 1 - \chi\varphi)Z + \chi = 0, \quad e_{1,2} = \frac{(\chi + 1 - \chi\varphi) \pm \sqrt{\Delta}}{2}$$

**Step 3) Complex and Real Areas on  $\varphi$ .**  $\Delta < 0$  on an interval of  $\varphi$  (depending on  $\chi$ )  $\Rightarrow e_{1,2}$  are true complex numbers, with common modulus  $|e| = \sqrt{\chi}$ .

**Step 4) Extension of the Complex Region & Constriction Coefficient.** In the complex region, the convergence criterion requires  $\chi < 1$ . Trying  $\chi = \kappa/|e_2|$ ,  $\kappa \in ]0, 1[$ , gives

$$\chi = \begin{cases} \frac{2\kappa}{\varphi - 2 + \sqrt{\varphi^2 - 4\varphi}} & \varphi > 4 \\ \kappa & \text{otherwise} \end{cases}$$

**Thesis setting:**  $\kappa = 1$ ,  $\varphi = \varphi_1 + \varphi_2 = 4.1 > 4 \Rightarrow \chi \approx 0.7298$ .

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# Backup: Full Numerical Results (Table 5.1)

| Func.     | Config.   | Mean best   | Med. best   | Std      | SR   | $t_{hit}$ | $SR^T$ | $t_{hit}^T$ |
|-----------|-----------|-------------|-------------|----------|------|-----------|--------|-------------|
| Sphere    | Can. PSO  | 2.63e-41    | 2.88e-43    | 9.00e-41 | 100% | 23.5      | 100%   | 44.5        |
|           | FIPS-Ring | 2.18e-6     | 5.59e-8     | 5.52e-6  | 100% | 36.0      | 100%   | 80.0        |
|           | FIPS-All  | 8.16e-3     | 2.07e-3     | 1.27e-2  | 100% | 38.0      | 20%    | 363.5       |
| Ackley    | Can. PSO  | $\approx 0$ | $\approx 0$ | 0        | 100% | 23.0      | 100%   | 47.0        |
|           | FIPS-Ring | 2.86e-3     | 5.41e-5     | 9.85e-3  | 100% | 39.0      | 95%    | 70.0        |
|           | FIPS-All  | 9.12e-2     | 7.46e-2     | 7.40e-2  | 100% | 31.5      | 20%    | 82.0        |
| Rastrigin | Can. PSO  | 0           | 0           | 0        | 100% | 32.0      | 100%   | 55.5        |
|           | FIPS-Ring | 2.67e-3     | 3.06e-8     | 1.03e-2  | 100% | 79.0      | 85%    | 185.0       |
|           | FIPS-All  | 1.66e-2     | 1.52e-3     | 6.03e-2  | 95%  | 64.0      | 35%    | 122.0       |

$N_{runs} = 20$ .  $SR/SR^T$ : success rate under  $\delta/\delta_{tight}$ .  $t_{hit}$ : median iterations to first success (successful runs only). Ackley Can. PSO reported as  $-4.441 \times 10^{-16}$ : machine precision, numerically  $\equiv 0$ .

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## Backup: The Five Neighborhood Topologies (Mendes et al.)

| Topology      | Avg. Distance | Diameter | Distribution Sequence                          |
|---------------|---------------|----------|------------------------------------------------|
| All           | 1.00          | 1        | $\langle 19 \rangle$                           |
| Ring          | 5.26          | 10       | $\langle 2, 2, 2, 2, 2, 2, 2, 2, 2, 1 \rangle$ |
| Four Clusters | 2.26          | 3        | $\langle 4.6, 4.8, 9.6 \rangle$                |
| Pyramid       | 2.04          | 4        | $\langle 5.4, 7.8, 5.4, 0.4 \rangle$           |
| Square        | 2.32          | 4        | $\langle 4, 7, 6, 2 \rangle$                   |

Statistics measured on  $N = 20$  particles (Mendes et al., 2004). Ring has the largest diameter (slowest information spread); All has diameter 1 (instant, full spread) but performed poorly in their experiments — consistent with the FIPS-All stagnation observed in this thesis.

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## Backup: FIPS Variants and Self-Inclusion

- **FIPS** (used in this thesis): unweighted,  $W(k) = 1$  for all neighbors.
- **wFIPS**:  $W(k) \propto$  fitness of neighbor  $k$ 's personal best.
- **wdFIPS**:  $W(k) \propto$  distance to the particle being updated.
- **Self** / **wSelf**: the particle's own best is explicitly included in  $N_i$ , receiving half the total weight by construction.

Mendes et al. found several *self-excluding* (“U-”) topologies performed as well as or better than their self-including counterparts — suggesting a particle's own history may already be reflected in its current position and velocity. This thesis uses only self-inclusive, unweighted FIPS (Ring, All).

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## Backup: Complete List of Limitations (§6.2)

- Restricted to a two-dimensional search space
- Only unweighted FIPS; only Ring and All topologies tested
- Symmetric initialization (vs. Mendes' asymmetric design)
- Descriptive statistics only — no formal hypothesis testing
- Extra benchmark functions (Griewank, Rosenbrock, Schaffer f6) shown but not used in the comparison
- No velocity clamping (constriction factor only; not compared against inertia weight + clamping)
- Only self-inclusive FIPS configurations (no self-excluding “U-” variants)
- Tolerance thresholds  $\delta, \delta_{\text{tight}}$  not systematically sensitivity-tested

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- D. Wang, D. Tan, L. Liu, "Particle Swarm Optimization Algorithm: An Overview," *Soft Computing*, vol. 22, 2017.
- R. Mendes, J. Kennedy, J. Neves, "The Fully Informed Particle Swarm: Simpler, Maybe Better," *IEEE Transactions on Evolutionary Computation*, vol. 8, no. 3, 2004.
- M. Clerc, J. Kennedy, "The Particle Swarm – Explosion, Stability, and Convergence in a Multidimensional Complex Space," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 1, 2002.
- J. Kennedy, R. Eberhart, "Particle Swarm Optimization," *Proceedings of ICNN'95 – International Conference on Neural Networks*, 1995.
- V. Lange, M. Schmitt, R. Wanka, "Towards a Better Understanding of the Local Attractor in Particle Swarm Optimization: Speed and Solution Quality," Springer, 2014.